

Open Source for Regulated Manufacturing

GMP-Ready IIoT Data Logging with Apache IoTDB

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The Pressure

Connected Manufacturing Meets Regulation



The Shopfloor Is Becoming a Data Platform



The Core Tension

- Manufacturing needs more real-time data.
- GMP requires trustworthy and inspectable records.
- AI and analytics increase the demand for structured, contextualized data.
- Legacy shopfloor architectures are often fragmented and opaque.

The future shopfloor needs data platforms that are both scalable and inspectable.

Regulatory Pressure Is Expanding

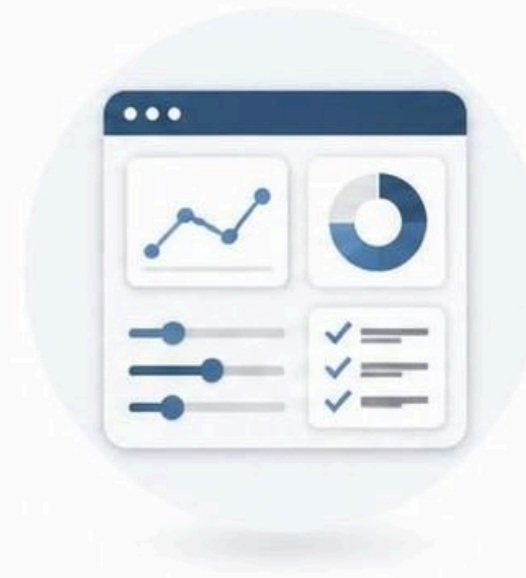


Two Questions Regulators Ask



**Is the
record
trustworthy?**

SECURITY • INTEGRITY • TRUST



**Is the
whole
system
controlled?**

GOVERNANCE • OVERSIGHT • CONTROL

The Regulatory Lens



FDA vs. EU in 60 Seconds

FDA 21 CFR Part 11



Electronic records
Accurate, reliable, and readily available.



Secure signatures
Unique to one individual and linked to their records.



Trusted paper-equivalent recordkeeping
Electronic records equivalent to paper records.



Trustworthy records and signatures.



EU Annex 11



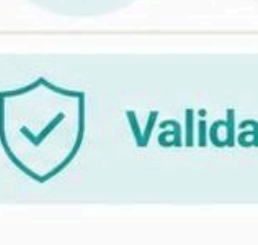
Validated computerized system
Fit for intended use and validated throughout.



Lifecycle control
Requirements managed from design to retirement.



Governance
Clear roles, responsibilities, and oversight.

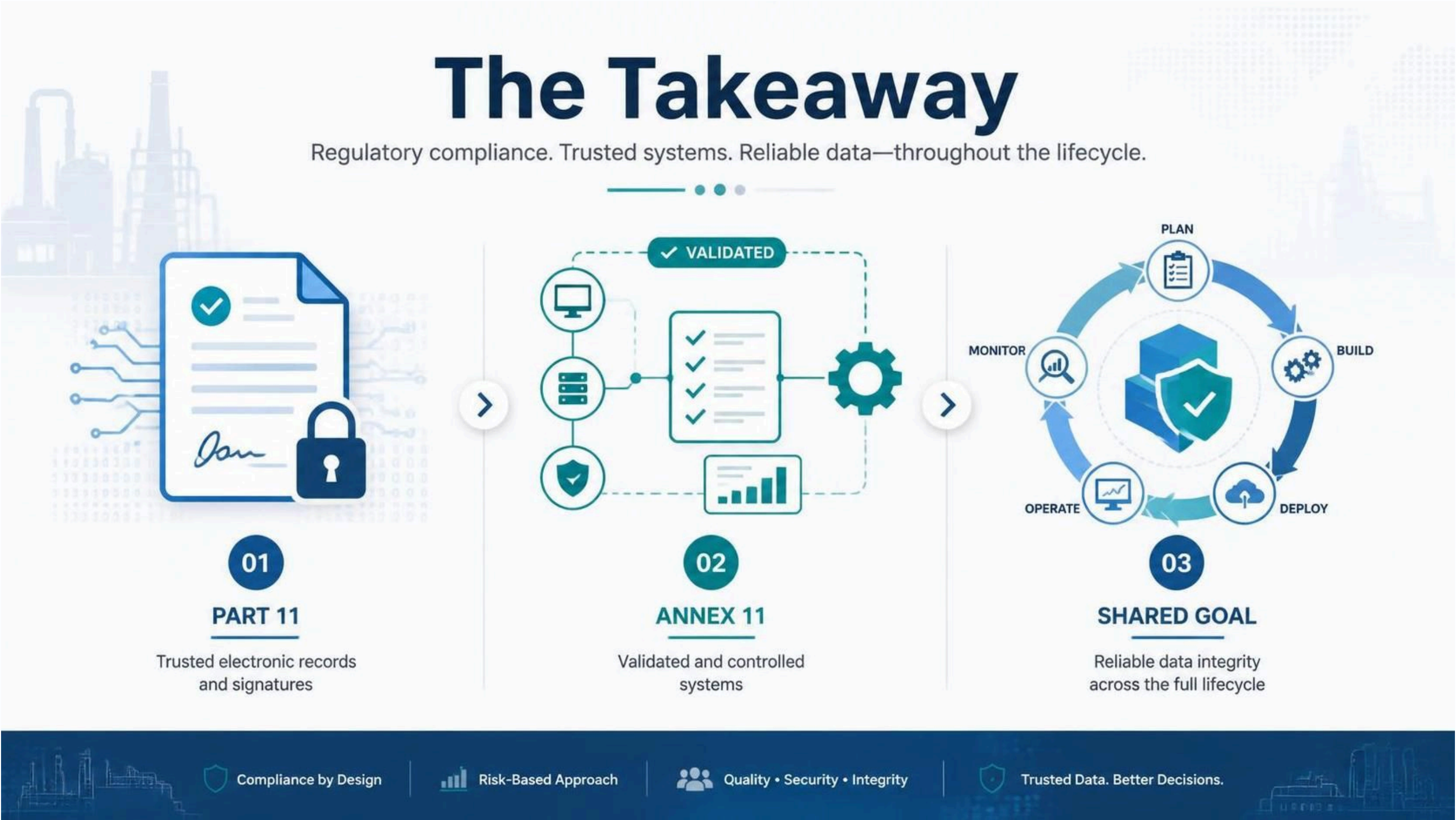


Configuration management
System changes controlled, tracked, and documented.



Validated and controlled systems.

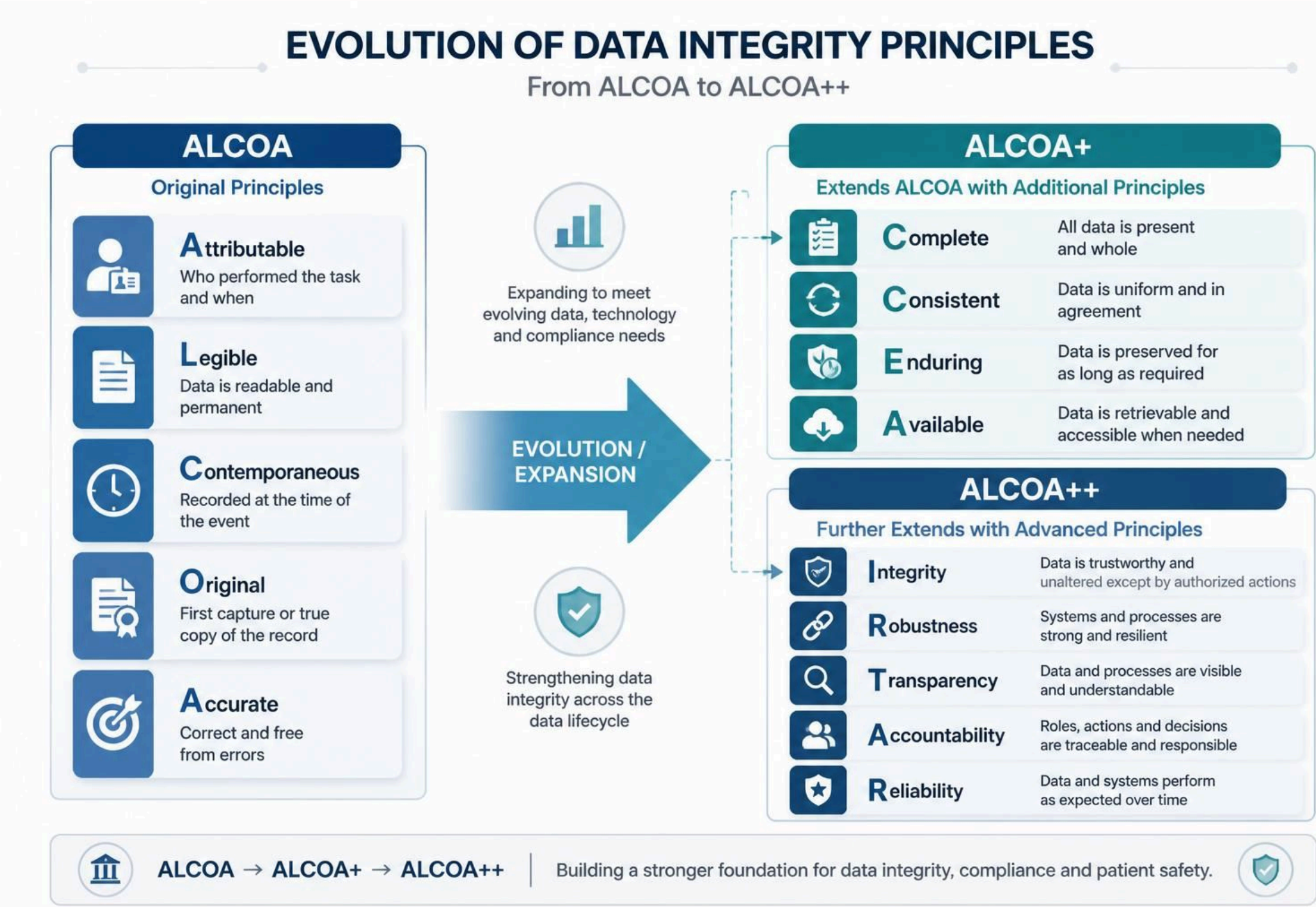
The Shared Compliance Goal



Data Integrity Becomes Architecture



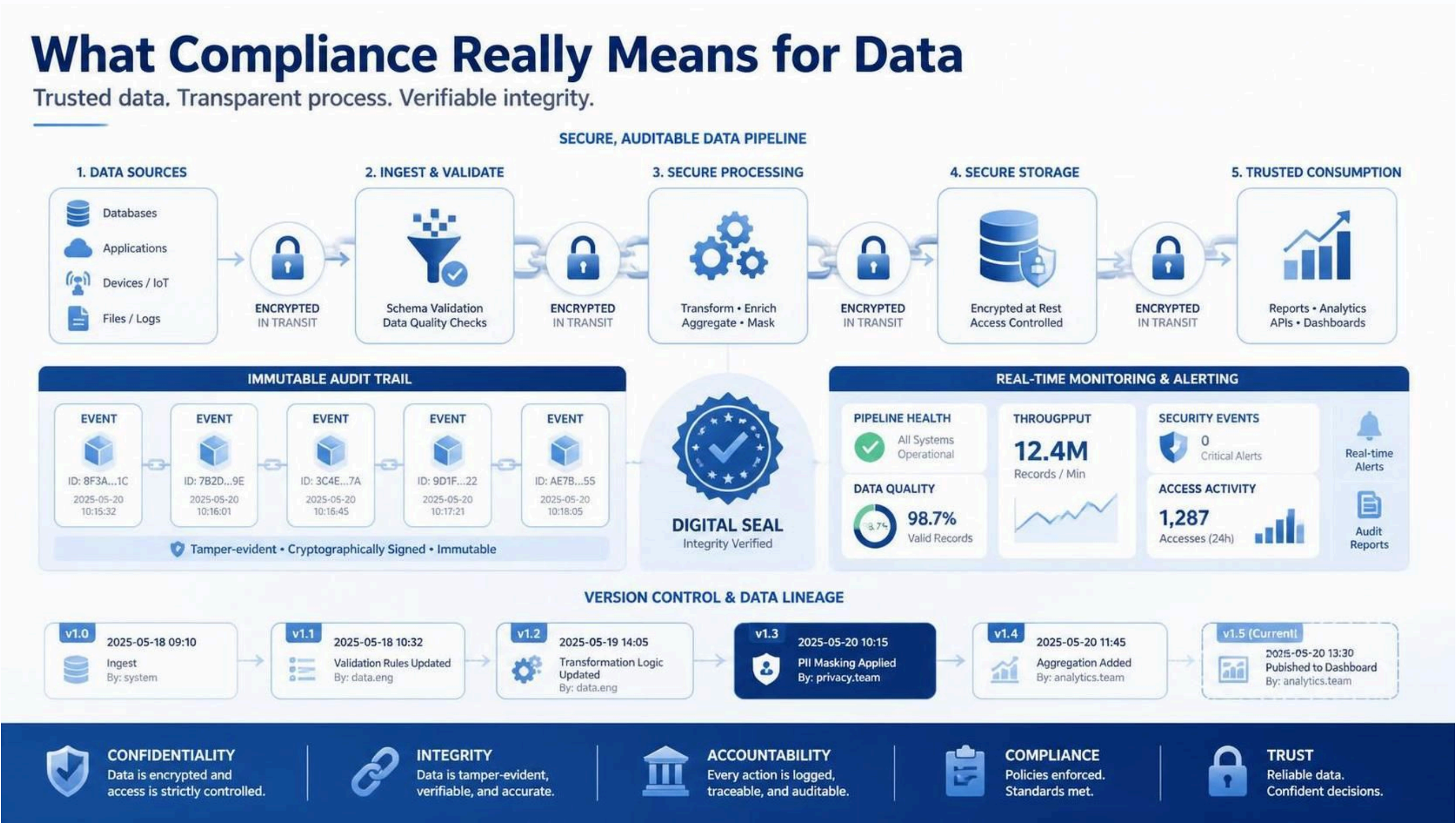
ALCOA++ as the Practical Lens



ALCOA++ Translated to IIoT

For IIoT, ALCOA++ is not a documentation checklist. It is an architecture requirement.

Attributable: Every reading is linked to a machine, sensor, gateway, service, and timestamp.	Contemporaneous: Data is captured immediately at the machine event.
Original: Raw values are preserved before any transformation or aggregation.	Complete: Retries, gaps, corrections, and synchronization events remain traceable .
Enduring: Records survive retention periods , infrastructure changes, and recovery scenarios.	Available: Authorized users can retrieve data at any time for operations, review, or inspection.
Transparent: Processing logic, configuration, and data movement are auditable.	Accountable: System and user actions are tied to responsible identities .



What Can Go Wrong on the Shopfloor?

Network outage during production	Duplicate readings after retry
Missing or inconsistent timestamps	Unauthorized data access
Silent data loss during synchronization	Uncontrolled transformation logic
Retention or backup failure	Configuration drift between environments

A GMP-ready IIoT architecture must be designed for failure, not only for normal operation.

Why Open Source?



Why Open Source Helps Validation

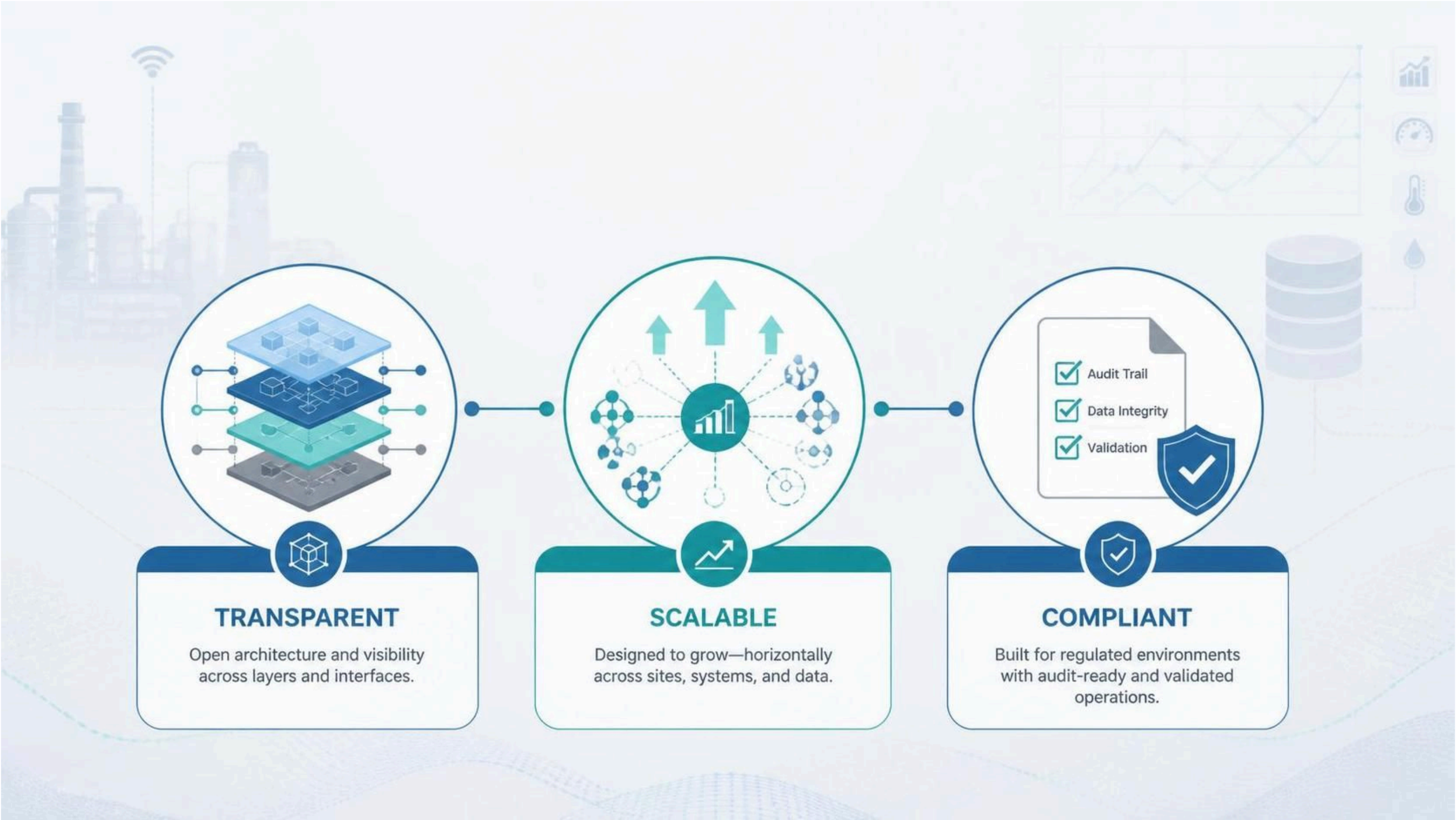


Open Source Is Not Automatically Compliant

- The software is not validated by default.
- The implemented system must be qualified and validated.
- Supplier and community governance still need assessment.
- Security patching and dependency management must be controlled.
- Documentation, testing, SOPs, and change control remain mandatory.

Open source is not a shortcut around GMP. It is a transparent foundation that can support validated architectures.

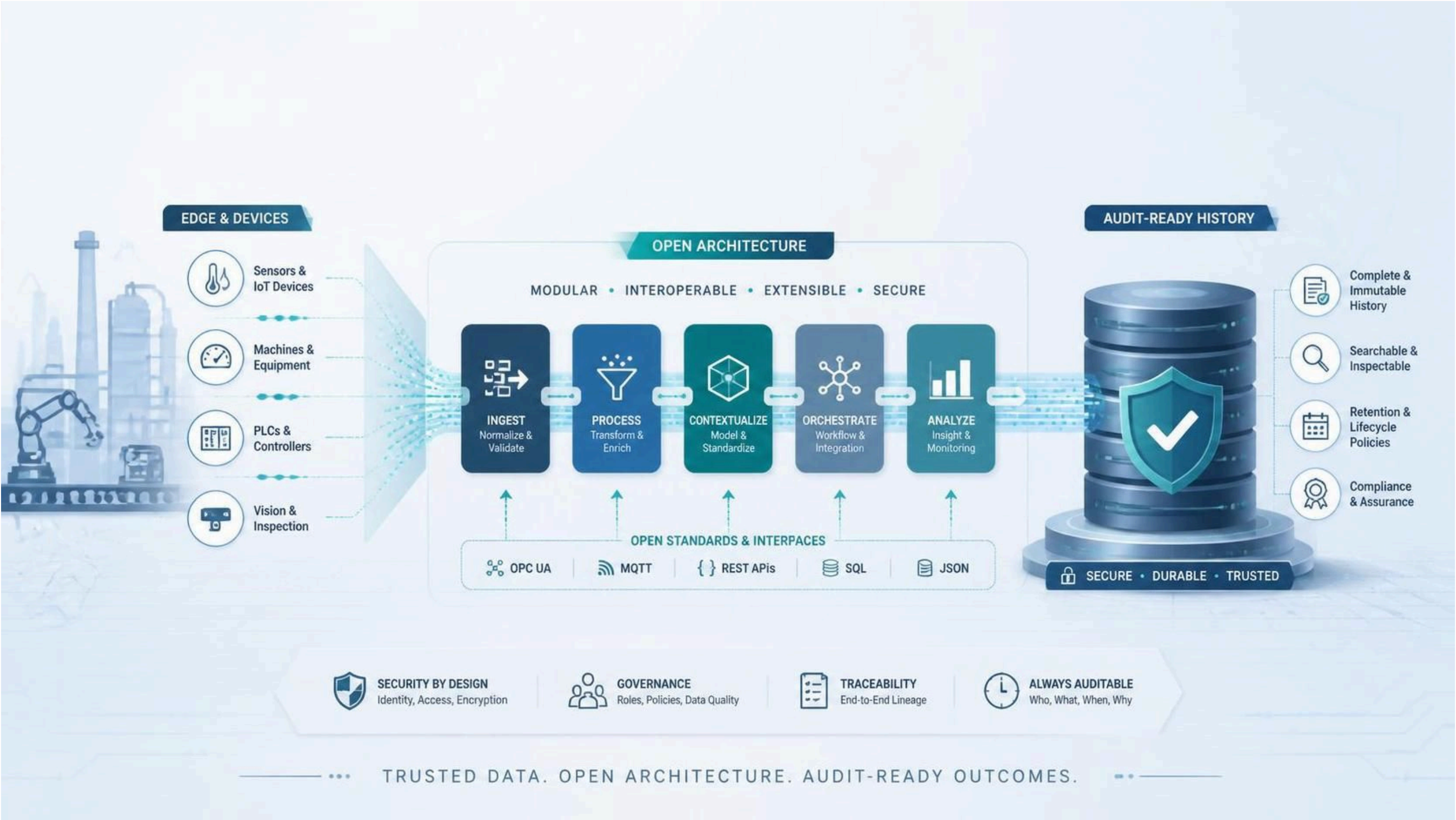
What to Remember About Open Source



From Open Architecture to Apache IoTDB



Open Architecture for Audit-Ready History



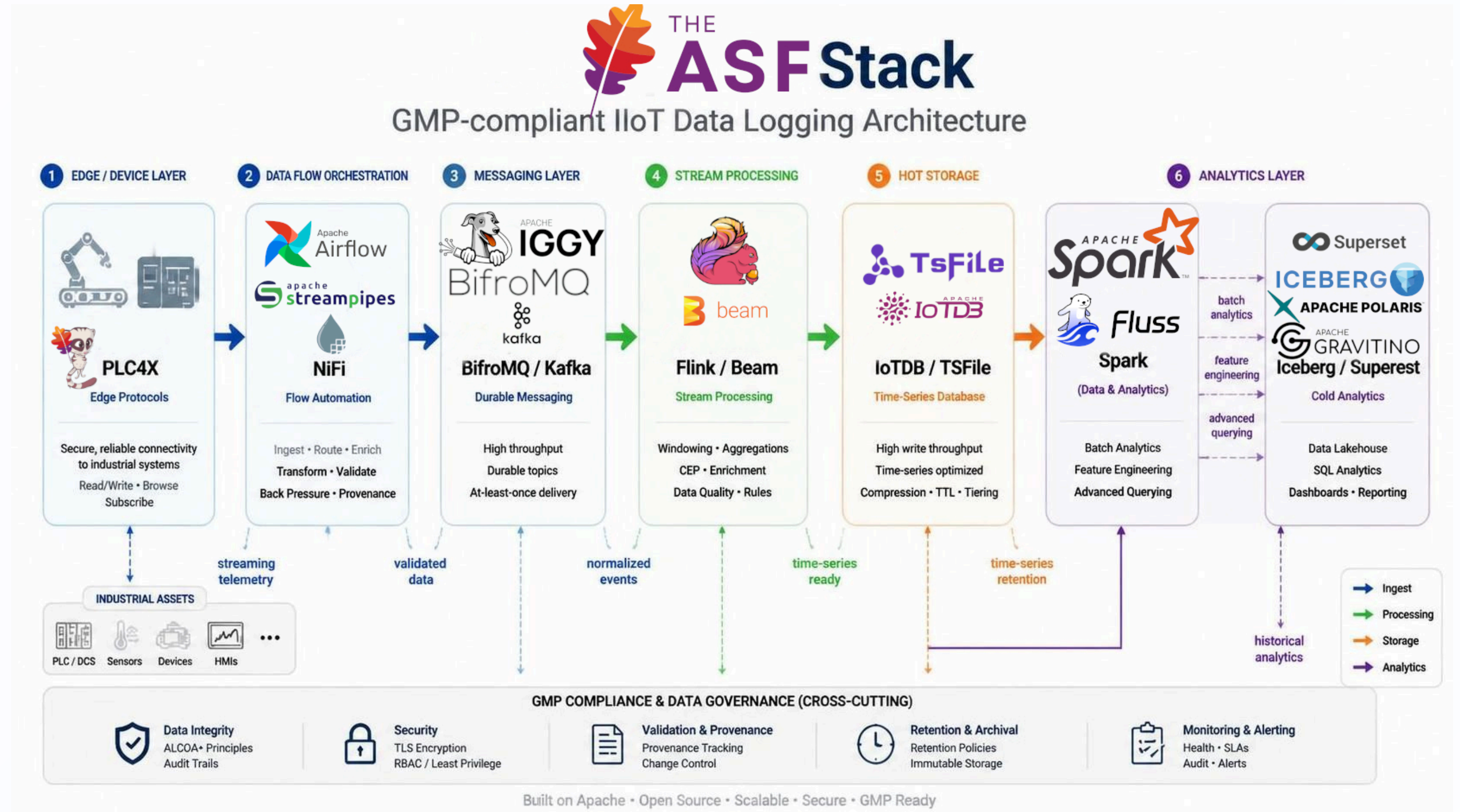
Example Intended Use

- Capture high-frequency machine and sensor readings.
- Buffer data locally when central connectivity is unavailable.
- Synchronize data into a historian without duplicates.
- Visualize current and historical data.
- Support traceability, operational resilience, and inspection readiness.

The goal is not to validate a database in isolation. The goal is to validate the implemented system for its intended use.

Apache-Based IIoT Data Logging

Reference Architecture



Apache IoTDB as the Historian Core

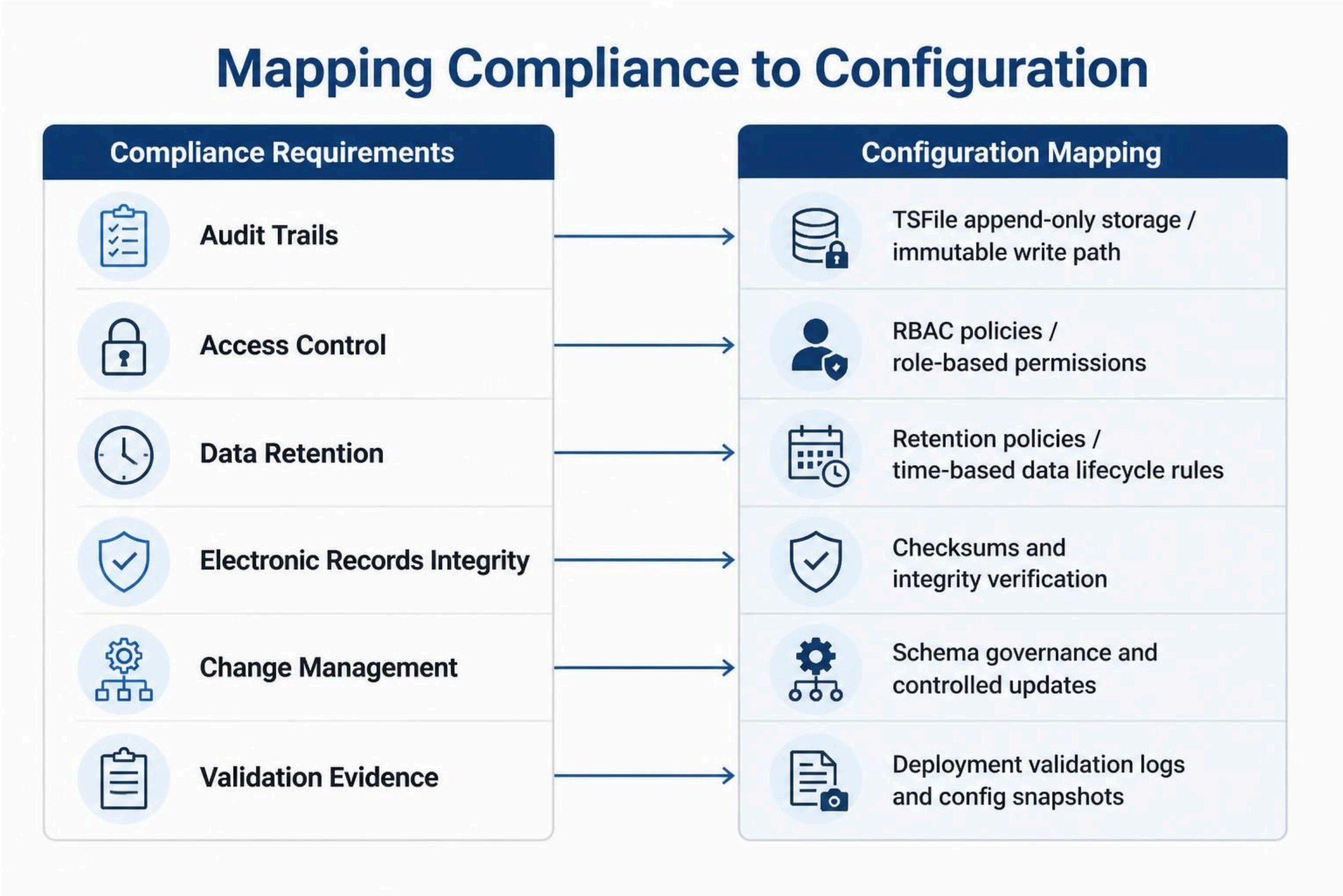
- Optimized for high-frequency industrial time-series data.
- Stores structured device and sensor measurements.
- Supports integration with authentication, authorization, encryption, monitoring, and retention controls.
- Can serve as the historian core inside a validated IIoT architecture.
- Works together with controlled ingestion, metadata, logging, and validation processes.

Apache TSFile as the Storage Foundation

- Time-series file format optimized for efficient storage and query.
- Supports append-oriented storage patterns.
- Helps separate raw time-series history from downstream analytics.
- Supports reproducibility and inspection when combined with controlled storage, access, and retention design.
- Can be paired with analytical formats such as Apache Iceberg.

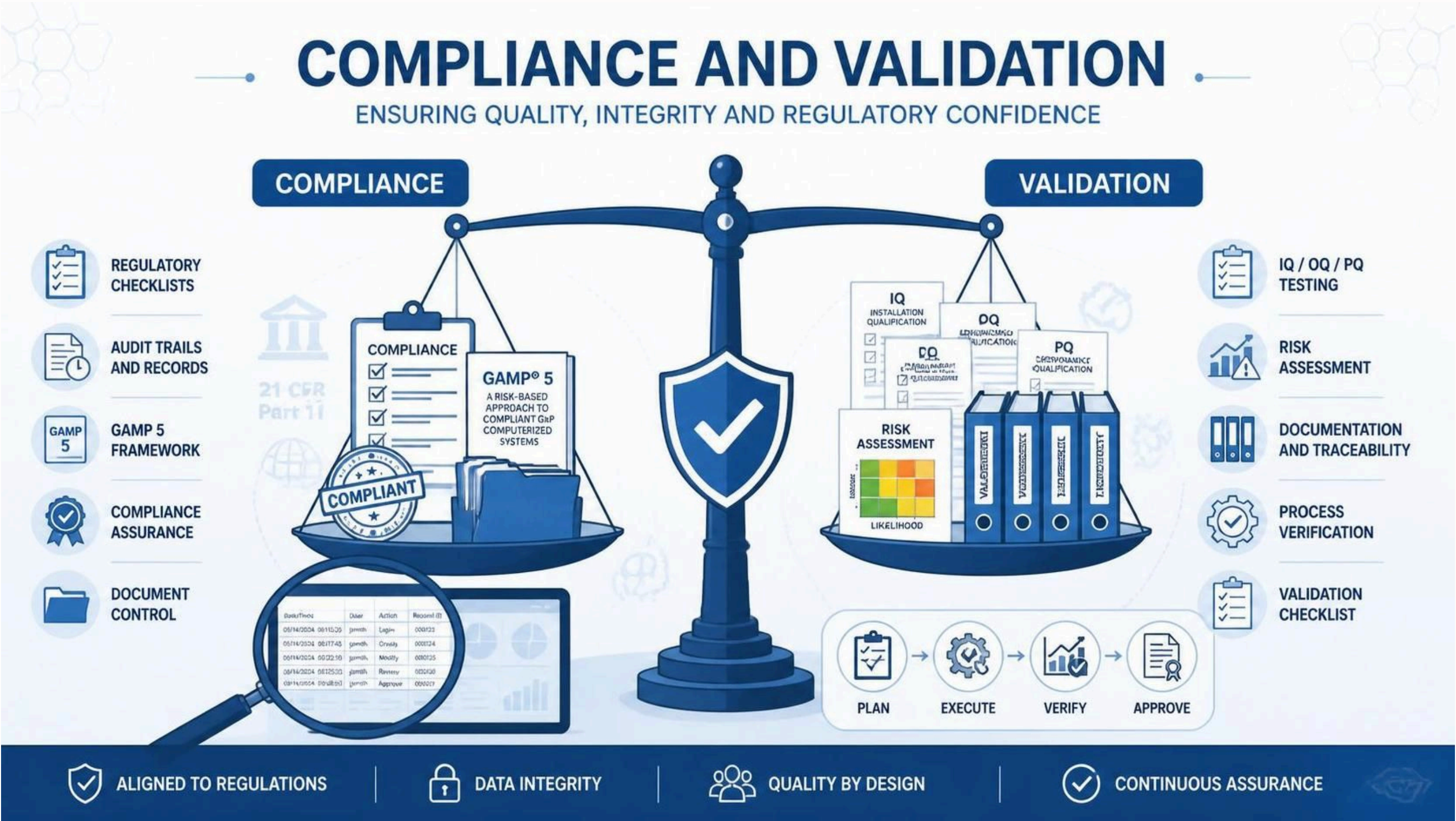
Mapping Compliance

Expectations to Technical Controls



From Architecture to Validation Evidence





Validation Starts with System Boundaries

- What is GMP-relevant?
- What is the regulated record?
- Which components create, modify, store, or display the record?
- Which integrations are in scope?
- Which users, services, and systems can access the data?
- What is the intended use?
- What is explicitly out of scope?

Risk-Based Validation with GAMP 5

1. **Intended Use**: Define what the system is expected to do.
2. **Risk Assessment**: Identify patient, product quality, and data integrity risks.
3. **User Requirements**: Define business, quality, and operational requirements.
4. **Design Specifications**: Translate requirements into controlled architecture and configuration.
5. **IQ / OQ / PQ**: Prove installation, operation, and performance.
6. **SOPs and Training**: Ensure controlled and repeatable use.
7. **Monitoring and Review**: Verify ongoing compliant operation.
8. **Change Control**: Manage updates without breaking validated state.

What Good Testing Looks Like

Test	What it proves	IIoT example
IQ	The system is installed repeatably and correctly.	Correct versions, configuration, certificates, storage paths, deployment scripts.
OQ	The system functions as specified.	RBAC, retention, backup, synchronization, failover, audit logging, monitoring.
PQ	The system performs under realistic production conditions.	Peak sensor load, network outage, replay, idempotent sync, long-running ingestion.

Main Elements of 21 CFR Part 11

21 CFR Part 11 requirements for trustworthy electronic records and signatures.



VALIDATION

Ensure systems are fit for intended use and consistently perform as expected.



SECURITY

Protect records and signatures to ensure confidentiality, integrity, and availability.



RECORD RETENTION

Retain records for the required period and ensure accurate, readable reproduction.



AUDIT TRAILS

Create secure, computer-generated audit trails that record time-stamped actions and changes.



ELECTRONIC SIGNATURES

Use electronic signatures that are unique to each individual and linked to their records.



TRAINING

Ensure personnel are trained on Part 11 requirements and their role in compliance.

The Demo: Resilience and Traceability



Demo Scenario

Network Outage Without Data Loss

- A production line continues generating readings.
- The central historian is temporarily unavailable.
- The local system continues accepting data.
- Readings are buffered safely.
- Synchronization resumes when connectivity returns.
- Duplicate records are avoided through idempotent behavior.
- Historical data becomes visible in Apache IoTDB.

The demo proves continuity, traceability, and recoverability under realistic shopfloor conditions.

Live Demo: Hybrid IoT Data Logging and Visualization

- Goal: show the path from ingestion to buffering to synchronization.

Demo Steps

Step	Action
1	Log in as <code>operator</code> with <code>operator</code> .
2	Send sample readings to <code>/ingest</code> or use the ingestion helper.
3	Open the dashboard and observe the local buffer chart.
4	Trigger <code>Sync to IoTDB</code> and watch the live SSE progress stream.
5	Verify that synchronized historical data appears in the IoTDB chart.

Conclusion



From Edge Data to Audit-Ready History



Final Takeaways

- Regulated manufacturing does not just need more data.
- It needs data that can be trusted, inspected, and preserved.
- Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture.
- Apache technologies provide modular building blocks from edge to analytics.
- Validation depends on intended use, risk, evidence, procedures, and lifecycle control.
- Open source supports transparency, scalability, and validatability.

Open source is not a shortcut around validation. It is a foundation for transparent, scalable, and inspectable manufacturing data platforms.

Q&A

- Thank you.
- Projects:
 - Apache IoTDB: <https://iotdb.apache.org/>
 - Apache PLC4X: <https://plc4x.apache.org/>
 - Apache StreamPipes: <https://streampipes.apache.org/>
- Contact:
 - Lukas Ott <otluk@apache.org>